rendered in the Cognite Data Fusion™ web-based digital twin platform is shown in figure 2.12.

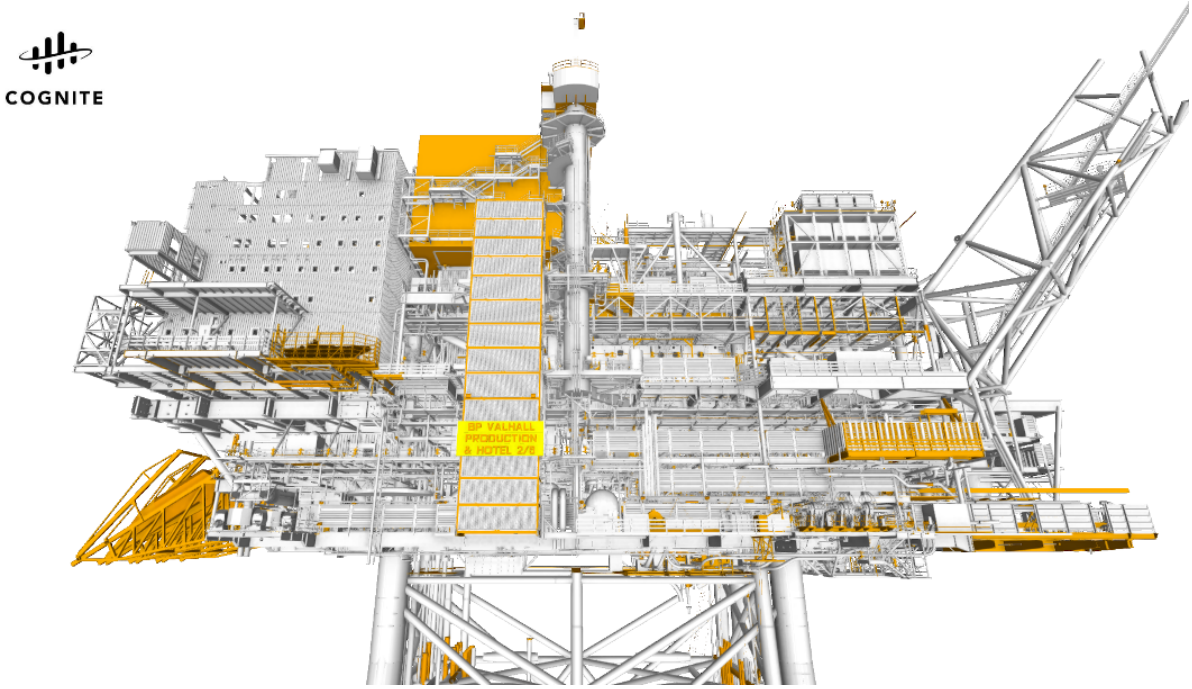


Figure 2.12 A 3D model derived from a CAD model of the Valhall production platform—a complex industrial facility located in the North Sea—as pictured in Cognite Data Fusion™, an industrial digital twin platform. Image © Cognite™ (https://www.cognite.com/en), reproduced with permission.

POINT CLOUDS

A point cloud is a collection of often millions or billions of points in 3D space, with each point having spatial co-ordinates (x,y,z) and color, timestamp, and intensity information. The huge number of discrete points rendered in 3D space collectively represent the geometric shape of physical objects at the time they are captured. This remarkably simple data structure, generated with laser or light scanning, or structure from motion, is a powerful mechanism for capturing and recreating physical objects within a digital twin, particularly large scale environments like buildings, infrastructure, and terrain.

Point clouds are particularly useful for tracking changes over time (with the appropriate equipment, laser scans can be performed quickly), and identifying wear or damage, particularly with complex or irregularly shaped